

Data Quality Analysis Report

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Executive Summary

This report analyzes data quality across 2 files containing 25,270 total records. The analysis identified 76,404 validation errors. **Key Recommendations:** • Address mandatory field completeness issues at the source • Implement data validation workflows before processing • Review critical validation errors for immediate remediation

File Quality Scores

File Name	Mandatory Complete	Total Records	Errors	Quality %
Activity_Online_Course.xlsx	100.0%	23,579	73,163	55.6%
Activity_Curriculum.xlsx	100.0%	1,691	3,241	42.5%

Error Summary

Error Type	Count	Severity	Recommended Action
TYPE_COMPATIBILITY	47,158	Critical	Workshop: incompatible types. Resolve compatibility.
ENCODING_ISSUE	27,824	High	Source: invalid characters. Fix character encoding.
HETEROGENEOUS_TYPE	644	High	Workshop: mixed data types. Standardize approach needed.
LEADING_SPACES	579	Medium	Transform: leading whitespace. Auto-fix with trim.
TRUNCATION	199	High	Source: field exceeds length limit. Adjust target or truncate data.

Cross-File Referential Integrity Analysis

This section analyzes referential integrity between files using dynamic relationship discovery from Neo4j field mappings. Critical integrity issues can impact data migration success.

Activities Domain Integrity Analysis

Business rule-based orphan detection for Activities domain relationships. This analysis identifies ILT courses and sessions that lack valid activity references.

Medium Priority Issues (3 relationships)

Relationship	Integrity %	Orphaned Records
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Activities.ClassCode → Activities.Activity Code (report-only)	0.0%	451.0
Activities.ClassCode → Activity Links.Child ActivityCode (report-only)	0.0%	451.0
Activities.SessionParts.ClassCode → Activity Links.Child ActivityCode (report-only)	0.0%	1.0

Integrity Summary

Metric	Value
Total Relationships Analyzed	3.0
Critical Integrity Issues	0.0
Total Orphaned Records	903.0

Detailed Integrity Analysis

Critical Issues (3 relationships)

Source → Target	Key Field	Integrity %	Orphaned Records	Business Impact
Activities.ClassCode → Activities.Activity Code (report-only)	ClassCode→Activity Code	0.0%	451.0	Medium: 451/451 records with integrity issues - impacts data quality
Activities.ClassCode → Activity Links.Child ActivityCode (report-only)	ClassCode→Child ActivityCode	0.0%	451.0	Medium: 451/451 activity references invalid - impacts reporting completeness
Activities.SessionParts.ClassCode → Activity Links.Child ActivityCode (report-only)	ClassCode→Child ActivityCode	0.0%	1.0	Medium: 1/1 activity references invalid - impacts reporting completeness

Cross-File Error Patterns (Legacy Analysis)

Error Pattern	Affected Files	Total Impact	Recommended Action
ENCODING_ISSUE	Activity_Online_Course.xlsx, Activity_Curriculum.xlsx	27,824 records	Source: invalid characters. Fix character encoding.
TRUNCATION	Activity_Online_Course.xlsx, Activity_Curriculum.xlsx	199 records	Source: field exceeds length limit. Adjust target or truncate data.